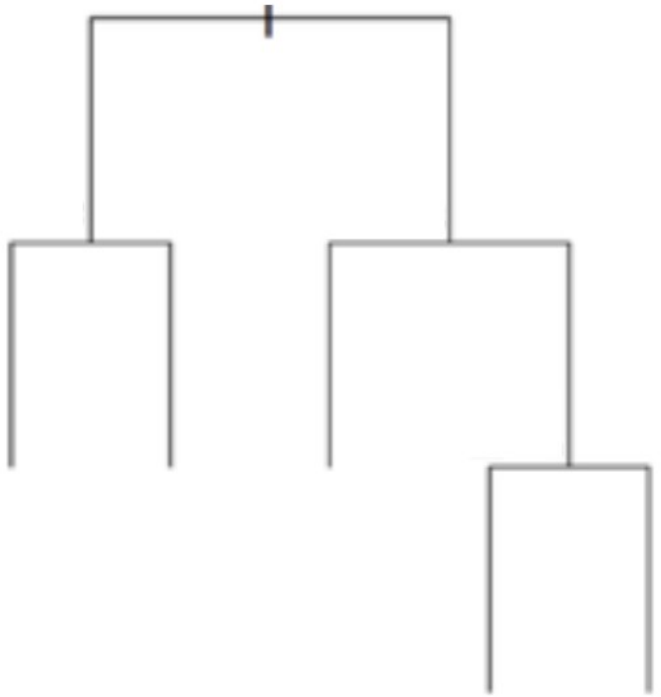
Lecture 13: Trees

Trees vs trees

- “Trees” are common
  - You have likely heard about trees as the things growing outside
  - You may have heard about decision trees
  - You have probably not heard of the 2014 Slovenian film
  - You have also likely heard about computer science trees as a data structure
    - The trees that we will discuss today are related to, *but quite distinct from*, these trees

Machine learning trees

- Represented in a similar way to how phylogenetic trees are represented in biology



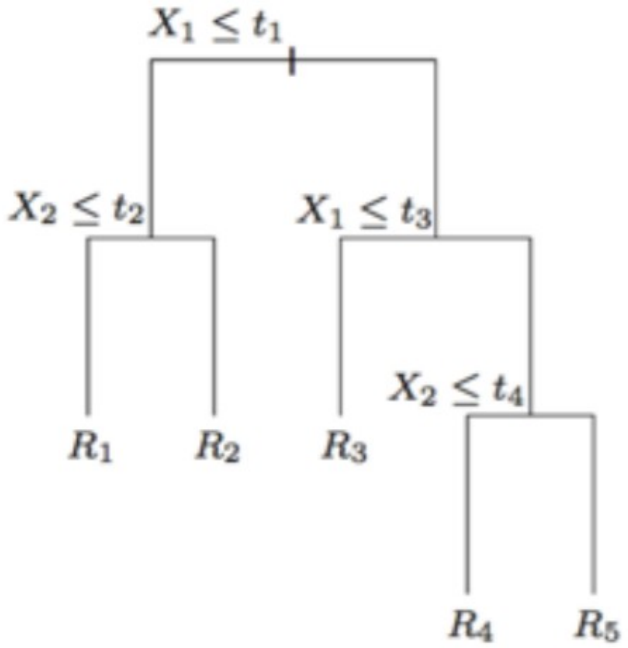
Machine learning trees

- Represented in a similar way to how phylogenetic trees are represented in biology
- Built as binary trees
- Data are stored at leaves with split points along axes

- How do you split? Greedily
- When do you stop? When points in leaves <= m

Learning *decision* trees

- Start with an empty decision tree
- Split on the next best feature/attribute
  - That is, split on $\text{argmax}_i(X_i) = \text{argmax}_i H(Y) - H(Y|X_i)$
- Recurse
- Prune



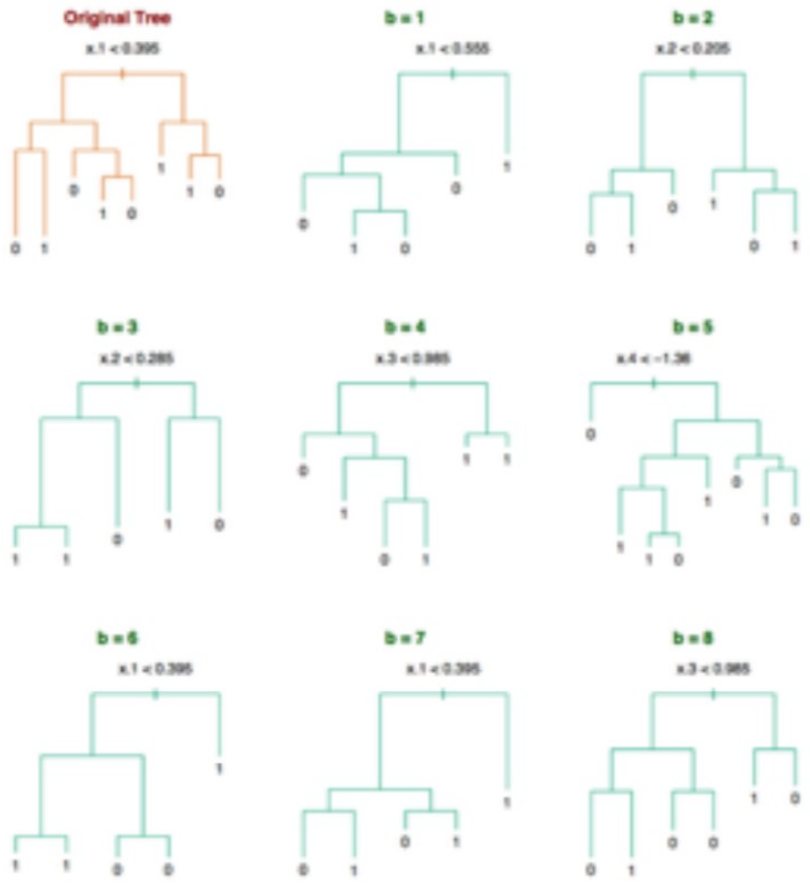
Trees

- Have low bias, high variance
- Deal quite well with categorical variables
- Intuitive & easily understandable
- Some theoretical guarantees (that we won’t go into)

Random forests

- Tree methods have low bias but high variance
- One way to reduce this variance is to generate a lot of related trees, assign weights to these trees, then average them
- This is “bagging” or “**b**ootstrap **agg**regat**ing**”

Random forests



Random forests

- To grow a random forest, continue until you reach the desired number of nodes:
  - Select m variables at random from among p variables
  - Pick the best variable (split point) among the m
  - Split the node into two daughter nodes
- Output the ensemble of trees

- This can be done for regression (mean tree) of classification (mode tree)

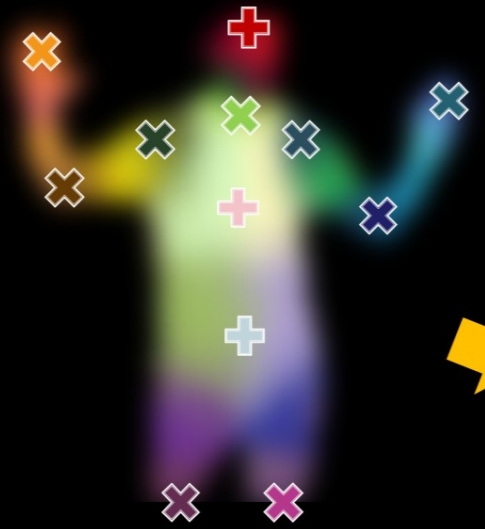
The Kinect pose estimation pipeline



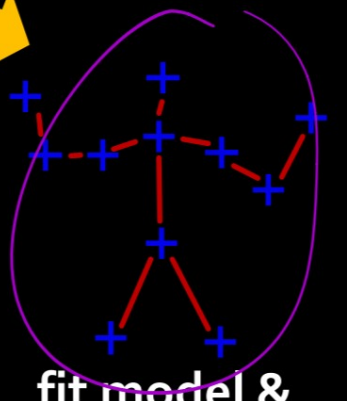
capture
depth image &
remove bg



infer
body parts
per pixel



cluster pixels to
hypothesize
body joint
positions

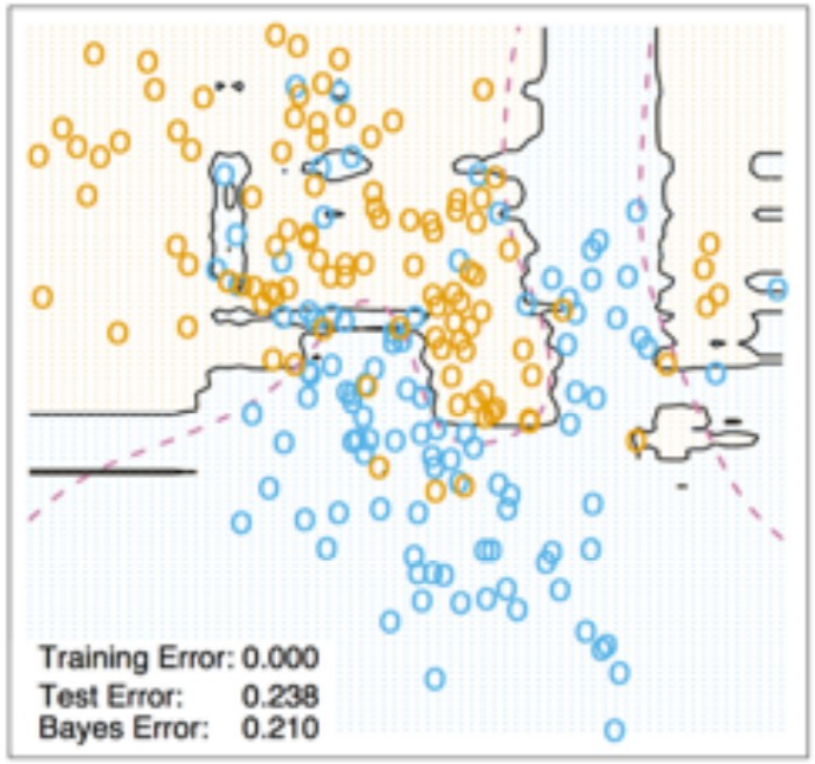


fit model &
track skeleton

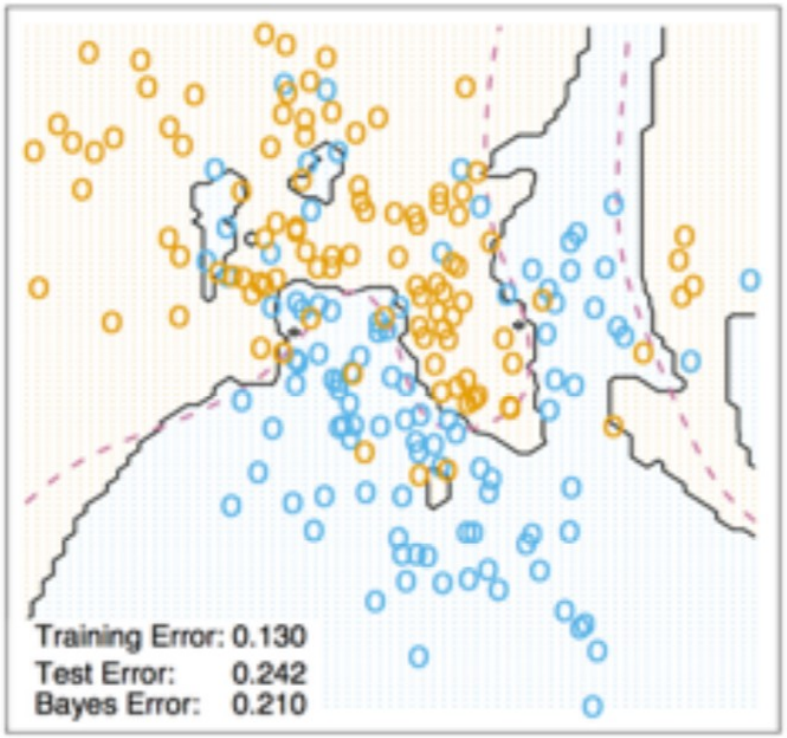
https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/CVPR20201120-20Final20Video.mp4

Random forest vs nearest neighbor

Random forrest



3 nearest neighbor



Random forests

- Have low bias, low variance (corrects decision trees' habit of overfitting)
- Deal with categorical variables well
- Not that intuitive or interpretable
- Some theoretical guarantees
- Can still overfit (although typically less likely than a single tree)

Ways to fix (or minimize) this: Bagging and Boosting next time